Supplementary material

Every table in this document is generated from a committed artifact by `scripts/build_supplement.py`. No figure is transcribed by hand.

Manuscript source commit `8780c72828d8`: the state of the manuscript and analysis sources from which this supplement was generated. This is a coauthor-review draft in a public repository. The code, protocol, design, frozen factor models and confirmatory analysis are pinned by the existing tags of S1; the later analyses, their scripts and this text are on the revision branch and will be pinned by the manuscript-package tag, still to be created after the authors approve the text (S17 names it as planned).

Protocol tag `xgboost-hpo-protocol-v3`. Results tag `xgboost-hpo-confirmatory-results-v1`.

S1. Protocol lineage and tags

tag	commit	what it records
`v0.1.0-dissertation`	`67d9fe50`	the historical implementation this study reconstructs
`xgboost-hpo-protocol-v1`	`355c2904`	the specification frozen before the pilot
`xgboost-hpo-protocol-v2`	`c50c380b`	the confirmatory protocol after pilot Stage A
`xgboost-hpo-protocol-v3`	`9b15ba7c`	the final frozen protocol, before the confirmatory campaign
`xgboost-hpo-confirmatory-results-v1`	`a4f20edd`	the first adversarially verified analysis

The protocol tag records what was specified before the confirmatory campaign began; the results tag records the first verified analysis. Neither was moved.

S2. Amendment chronology

The protocol carries 23 numbered amendments, each recording what caused it and whether any arm result had been observed when it was made; the ledger records that none had. It also records two process failures found by review rather than by the author, and four engineering defects caught by pre-freeze smoke runs. The last of these runs, on 2026-09-14, completed one full Spambase unit, every arm and the metrics stage, before amendments 21 (primary reference) and 22 (Spambase's role) and before the v3 tag; the ledger records that only its execution and accounting were inspected, and its outputs are not archived. The ledger is reproduced in full in `PROTOCOL_AMENDMENTS.md`. The corrections made after the campaign to two claims from the analysis are recorded in `analysis/verified_claims.json`, not in the ledger.

S3. Hyperparameter space

hyperparameter	lower	upper	type
`subsample`	0.05	1.0	continuous
`colsample_bytree`	0.05	1.0	continuous
`colsample_bylevel`	0.05	1.0	continuous
`learning_rate`	0.01	0.3	continuous
`max_depth`	3	18	integer
`gamma`	0.05	5.0	continuous
`n_estimators`	50	700	integer

All seven enter the design in coded units on $[-1, 1]$; integers are cast by `int(round(·))` at evaluation time, with ties rounded to even. The `max_depth` mid-level of 10.5 is therefore evaluated as 10, and the 22 design runs at that level (the 10 centre runs and the 12 axial runs of the other six factors) enter every replication's surface fits at that value, coded $-1/15$ rather than 0.

S4. Responses and objective construction

response	role	transform
`Accuracy_Mean`	quality	none
`Precision_Mean`	quality	none
`Recall_Mean`	quality	none
`Specificity_Mean`	quality	none
`RocAuc_Mean`	quality	none
`LogLoss_Mean`	quality	none
`Leaves_Mean`	cost	log1p

S4.1 Frozen factor models

dataset	fitting sample	audit rows used	quality weights	Kaiser	λ_3/λ_4
MAGIC	88 pilot design rows	0	0.5556, 0.4444	2	2.93
Spambase	88 pilot design rows	0	0.7749, 0.2251	2	2.10
Adult	88 pilot design rows	0	0.5470, 0.4530	1	1.34
Bank Marketing	88 pilot design rows	0	0.5875, 0.4125	1	1.06

One model per dataset, fitted on the 88 design rows only and applied to all 30 replications. The fitting rows are the pre-campaign pilot (Stage A) evaluations of the design: stratified, shuffled five-fold cross-validation on the 80% training part of one stratified 80/20 partition per dataset (seed 20260913), the partition of the S5 screening, drawn separately from the 30 confirmatory partitions. Each replication re-evaluates the same 88 configurations on its own partition and maps them through the frozen model. Because the pilot partition is a separate draw, its training part shares rows with every replication-level holdout, roughly four in five of them as two independent 80/20 draws imply. A replication-level holdout is unseen by arm training and selection within its replication; it is not external to the frozen objective mapping, which was fitted on 88 cross-validated metric summaries computed on the pilot's training part. That overlap is with the measurement definition, not with any model scored on the holdout: no model trained on a replication's own rows sees its holdout. The 78-point audit-only external construction is excluded from the fit (amendment 23): 64 of its 78 coded points are identical to the external validation set and the rest are the same axial runs.

Eigenvalues of the correlation matrix of the seven standardized responses Z on the 88 fitting rows; the cumulative percentage of the total variance of Z carried by the first k components, $100 \sum_{j \le k} \lambda_j / 7$; and the percentage of the variance of the transformed leaf count that the first k components reproduce. Varimax redistributes the $k = 3$ total among the rotated factors without changing it. The frozen model is applied unchanged to every replication, so these values are the same in every replication. They are descriptive: the component count is fixed at three for every dataset and was not chosen by them. From `analysis/factor_dimensionality.json`, written by `scripts/factor_dimensionality.py` after the campaign from the committed pre-campaign reference models.

dataset	λ_1	λ_2	λ_3	λ_4	cumulative % of variance, $k = 1$ / **2** / **3**	leaf-count variance reproduced, %, $k = 2$ / **3**
MAGIC	3.988	1.659	0.891	0.304	57.0 / 80.7 / 93.4	17.6 / 98.8
Spambase	4.455	1.092	0.865	0.412	63.6 / 79.2 / 91.6	68.2 / 84.0
Adult	4.775	0.851	0.701	0.522	68.2 / 80.4 / 90.4	78.3 / 98.1
Bank Marketing	4.664	0.853	0.625	0.588	66.6 / 78.8 / 87.7	75.3 / 89.1

S4.2 Rotated loadings and objective coefficients

For each frozen model: the oriented rotated loadings Λ_R of the seven canonicalized (minimization-oriented) responses on the leading quality factor (the one with the larger quality weight in S4.1), the second quality factor and the cost factor; the correlation of each response with f_{quality}; and the coefficients of the two objectives on the standardized responses Z (`Leaves_Mean` after its $\log(1+\cdot)$ transform). A loading is the Pearson correlation of a response with a factor score on the 88 fitting rows, so in the loading and correlation columns a positive entry means that the response tends to worsen as the score rises on those rows. The frozen model is applied unchanged to every other evaluation, where these correlations are not guaranteed to hold (S4.3). In the coefficient columns a positive entry means that, with the other responses held fixed, a worse value of that response raises the objective. The responses do not vary one at a time, so a coefficient describes the fixed linear map, not an effect of that response, and a coefficient and the loadings can differ in sign. Accuracy is, fold by fold, the prevalence-weighted mean of recall and specificity, so the coefficients on those three are not separately interpretable. Neither quality factor names the same quantity on two datasets. The coefficients are read from `analysis/factor_dimensionality.json`.

dataset	response	leading quality factor	second quality factor	cost factor	correlation with f_{quality}	f_{quality} coefficient	f_{cost} coefficient
MAGIC	Accuracy_Mean	+0.862	-0.433	-0.196	+0.402	+0.115	-0.015
MAGIC	Precision_Mean	-0.047	+0.965	+0.024	+0.566	+0.279	-0.128
MAGIC	Recall_Mean	+0.750	-0.614	-0.200	+0.202	+0.028	-0.008
MAGIC	Specificity_Mean	-0.167	+0.955	+0.145	+0.466	+0.243	-0.015
MAGIC	RocAuc_Mean	+0.933	-0.018	-0.161	+0.717	+0.262	-0.030
MAGIC	LogLoss_Mean	+0.898	+0.049	+0.061	+0.732	+0.275	+0.192
MAGIC	Leaves_Mean	-0.133	+0.115	+0.978	-0.032	+0.011	+1.006
Spambase	Accuracy_Mean	+0.894	+0.179	-0.348	+0.909	+0.195	-0.005
Spambase	Precision_Mean	+0.108	+0.988	-0.094	+0.379	+0.175	+0.119
Spambase	Recall_Mean	+0.917	+0.166	-0.311	+0.927	+0.209	+0.031
Spambase	Specificity_Mean	-0.922	+0.004	-0.145	-0.884	-0.302	-0.393
Spambase	RocAuc_Mean	+0.584	+0.051	-0.709	+0.575	-0.012	-0.426
Spambase	LogLoss_Mean	+0.885	+0.055	-0.392	+0.865	+0.159	-0.066
Spambase	Leaves_Mean	-0.129	-0.134	+0.898	-0.161	+0.178	+0.706
Adult	Accuracy_Mean	+0.776	+0.585	-0.180	+0.971	+0.189	+0.132
Adult	Precision_Mean	+0.582	+0.473	-0.134	+0.750	+0.146	+0.104
Adult	Recall_Mean	+0.861	+0.444	-0.216	+0.947	+0.182	+0.099
Adult	Specificity_Mean	-0.928	-0.124	+0.262	-0.794	-0.147	-0.026
Adult	RocAuc_Mean	+0.301	+0.791	-0.434	+0.737	+0.052	-0.290
Adult	LogLoss_Mean	+0.282	+0.935	-0.021	+0.813	+0.165	+0.197
Adult	Leaves_Mean	-0.242	-0.130	+0.951	-0.269	+0.165	+0.992
Bank Marketing	Accuracy_Mean	+0.731	+0.607	-0.235	+0.947	+0.209	+0.138
Bank Marketing	Precision_Mean	+0.636	+0.426	-0.015	+0.766	+0.232	+0.328
Bank Marketing	Recall_Mean	+0.866	+0.343	-0.316	+0.906	+0.196	+0.080
Bank Marketing	Specificity_Mean	-0.894	-0.121	+0.349	-0.801	-0.170	-0.030
Bank Marketing	RocAuc_Mean	+0.349	+0.595	-0.614	+0.627	-0.029	-0.480
Bank Marketing	LogLoss_Mean	+0.225	+0.943	-0.091	+0.726	+0.139	+0.105
Bank Marketing	Leaves_Mean	-0.298	-0.078	+0.893	-0.288	+0.189	+0.883

dataset	role-block mean, leading / second quality factor	Leaves_Mean on the cost factor	Spearman of f_{quality} with equal-weight quality badness, fitting rows
MAGIC	+0.538 / +0.150	+0.978	+0.929
Spambase	+0.411 / +0.240	+0.898	+0.243
Adult	+0.312 / +0.517	+0.951	+0.835
Bank Marketing	+0.319 / +0.466	+0.893	+0.896

The sign rule (Section 5.5) makes the two role-block means and the `Leaves_Mean` loading positive; it does not make every loading in a role block positive. The last column is the frozen orientation check of amendment 19 on the fitting rows; S4.3 evaluates it beyond them.

At the 0.5 threshold, 0 (MAGIC), 1 (Spambase), 13 (Adult) and 16 (Bank Marketing) of the 88 fitting rows predict no positive case in any fold, so their recall and precision are 0 and their specificity 1. All of them are among the 16 factorial design rows with `learning_rate` 0.01 and `n_estimators` 50 (Section 5.4).

S4.3 The quality composite beyond its fitting sample (post hoc)

Computed after the campaign by `scripts/composite_beyond_fitting_rows.py` from stored evaluations, with the frozen models; it changes no frozen number and fed no arm. Alignment is the frozen orientation check of amendment 19: the Spearman correlation of f_{quality} with the equal-weight mean of the six standardized canonicalized quality responses, which must be positive. Before the campaign it was evaluated on the 88-row fitting sample, and each replication recomputed it on its own design rows. Here it is also evaluated on the 386 configurations that the random-search baseline drew uniformly from the coded decision box in each replication. A reversal is a pair of configurations in which one is at least as good on all six quality responses and strictly better on one, but receives the larger (worse) f_{quality}; random-search pairs are formed within replications and pooled.

dataset	alignment, fitting sample	alignment, design rows (range over replications)	alignment, random-search rows: median (range)	replications with negative alignment	reversals, fitting sample (pairs)	reversal rate, random-search rows
MAGIC	+0.929	+0.889 to +0.945	+0.941 (+0.903 to +0.979)	0/30	0/905	<0.001
Spambase	+0.243	+0.188 to +0.310	-0.387 (-0.540 to -0.182)	30/30	1,045/1,311	0.751
Adult	+0.835	+0.809 to +0.850	+0.562 (+0.461 to +0.639)	0/30	150/615	0.262
Bank Marketing	+0.896	+0.873 to +0.916	+0.694 (+0.602 to +0.761)	0/30	149/421	0.146

S4.4 Per-replication refit: Tucker congruence

Declared in the frozen protocol (section 7.2; amendment 14) and computed after the campaign from stored outputs by `scripts/tucker_congruence_refit.py`. In each replication the factor model refitted on that replication's own 88 design rows is compared with the frozen model by the Tucker congruence of matched loading columns; the refit is never applied. The runner recorded congruences under a greedy column matching that does not resolve factor sign, so a negative value there can be an artifact of the matching. The optimal-assignment matching beside it is post hoc: columns are matched by maximal total absolute congruence and each matched column is oriented before its congruence is computed. A congruence of 0.95 or more is conventionally read as equivalence, and 0.85 to 0.95 as fair similarity.

dataset	greedy, as recorded: minimum absolute congruence	greedy: replications with a negative congruence	optimal (post hoc): minimum congruence	below 0.95	below 0.85	cost factor recovered
MAGIC	0.895	0 of 30	0.895	6 of 30	0 of 30	30 of 30
Spambase	0.918	0 of 30	0.918	2 of 30	0 of 30	30 of 30
Adult	0.948	0 of 30	0.948	1 of 30	0 of 30	30 of 30
Bank Marketing	0.351	4 of 30	0.751	10 of 30	4 of 30	30 of 30

S4.5 Front-level weighting sensitivity

Declared in the frozen protocol (section 11, item 1) and computed after the campaign from stored outputs by `scripts/weighting_sensitivity_fronts.py`. The protocol asks how often variance weighting and equal weighting of the two quality factors disagree about the returned front. Two fronts differ if their non-dominated sets differ by at least one member after deduplication at the evaluation tolerance, and if they differ in more than 20% of replications on a dataset, the conclusions for that dataset are reported as conditional on the weighting, in the abstract as well. The protocol did not fix how the equal-weight fronts are obtained or which arms the rule counts. Post hoc, each arm's stored revalidated candidates, and the CORE reference, are rescored with the equal-weight composite (no arm was re-solved), configurations with identical stored evaluations count as one member, and the rule is counted as met on a dataset if it is met for any of the four arms. The design-row rank agreement of the two composites (Spearman 0.987 to 0.991, Section 5.5) is a different estimand.

dataset	HISTORICAL-WS: fronts differ	WS-S: fronts differ	NBI-S: fronts differ	NBI-R: fronts differ	more than 20% for any arm
MAGIC	13 of 30	9 of 30	19 of 30	14 of 30	yes
Spambase	14 of 30	15 of 30	23 of 30	14 of 30	yes
Adult	13 of 30	17 of 30	14 of 30	18 of 30	yes
Bank Marketing	16 of 30	14 of 30	16 of 30	21 of 30	yes

The three identifying contrasts under the equal-weight composite, against the rescored CORE reference: median paired difference in the hypervolume ratio (second minus first), with W/T/L; descriptive and untested.

dataset	contrast	median difference	W/T/L
MAGIC	HISTORICAL-WS → WS-S	+0.0000	14/3/13
MAGIC	WS-S → NBI-S	+0.0395	27/0/3
MAGIC	NBI-S → NBI-R	-0.0776	1/0/29
Spambase	HISTORICAL-WS → WS-S	+0.0000	16/6/8
Spambase	WS-S → NBI-S	+0.0223	16/2/12
Spambase	NBI-S → NBI-R	-0.5934	5/1/24
Adult	HISTORICAL-WS → WS-S	-0.0012	6/2/22
Adult	WS-S → NBI-S	+0.2543	30/0/0
Adult	NBI-S → NBI-R	-0.0266	12/0/18
Bank Marketing	HISTORICAL-WS → WS-S	+0.0000	9/7/14
Bank Marketing	WS-S → NBI-S	+0.3205	27/0/3
Bank Marketing	NBI-S → NBI-R	-0.4389	7/0/23

S4.6 Leaf count and measured time

Declared in the frozen protocol (`protocol/OBJECTIVE_DEFINITIONS.md`, section 9, and `protocol/COST_OBJECTIVE_CLAIM_BOUNDARY.md`) and computed after the campaign from stored outputs by `scripts/leaf_time_association.py`: the association between the leaf-count objective and measured mean per-fold time over all 120 units. Before the campaign, raw total leaf count tracked measured per-fold time at Spearman 0.86 on 22 of the 88 MAGIC design configurations, and two timings of one configuration differed by a median 1.3% (`audits/cost_objective_selection.json`). Campaign timings were taken with several worker processes running in parallel (S17), so they carry load-dependent noise. The association is reported as measured; a weaker association is a finding about the proxy and changes no objective. Each cell is the median over units of the

per-unit Spearman correlation, with its range. Post hoc choices: all evaluations are every cross-validated fit stored for the unit, excluding holdout fits and the unmatched NSGA-II run, which lies outside the unit; design rows are the 79 distinct design configurations, because the ten replicated centre points were fitted once.

dataset	units	all evaluations: raw leaf count	all evaluations: f_{cost}	design rows: raw leaf count	design rows: f_{cost}
MAGIC	30	0.933 (0.919 to 0.943)	0.917 (0.899 to 0.930)	0.907 (0.897 to 0.915)	0.896 (0.886 to 0.904)
Spambase	30	0.957 (0.953 to 0.961)	0.919 (0.896 to 0.926)	0.935 (0.920 to 0.941)	0.820 (0.772 to 0.852)
Adult	30	0.871 (0.844 to 0.885)	0.852 (0.822 to 0.871)	0.870 (0.864 to 0.876)	0.795 (0.779 to 0.825)
Bank Marketing	30	0.878 (0.847 to 0.895)	0.851 (0.825 to 0.872)	0.894 (0.887 to 0.903)	0.729 (0.644 to 0.757)

S4.7 Cost-definition sensitivity

Declared in the frozen protocol (section 11, item 4): whether the primary cost objective and measured time disagree about the winning arm, reported as a frequency over replications. Computed after the campaign from stored outputs by `scripts/cost_definition_sensitivity.py`. The protocol did not fix the operationalization. Post hoc, the winning arm is the one of HISTORICAL-WS, WS-S, NBI-S and NBI-R with the largest CORE-relative hypervolume ratio; under the time definition the cost axis of every candidate and of the CORE reference is the measured mean per-fold time, with the quality objective unchanged. No arm was re-run, and the timing noise of S4.6 applies. The time axis is in seconds, untransformed, and arms whose ratios agree to within 10^{-12} share the win (listed as tied).

dataset	replications in which the winning arm differs	winners, leaf-count cost	winners, time cost
MAGIC	9 of 30	NBI-S 24, WS-S 4, HISTORICAL-WS 1, NBI-R 1	NBI-S 17, NBI-R 6, WS-S 5, HISTORICAL-WS 2
Spambase	17 of 30	NBI-S 14, WS-S 6, HISTORICAL-WS 3, NBI-R 3, HISTORICAL-WS/WS-S (tied) 2, HISTORICAL-WS/WS-S/NBI-S (tied) 2	NBI-S 15, NBI-R 5, HISTORICAL-WS 4, WS-S 3, HISTORICAL-WS/WS-S/NBI-S/NBI-R (tied) 2, HISTORICAL-WS/WS-S (tied) 1
Adult	16 of 30	NBI-S 19, NBI-R 11	NBI-R 16, NBI-S 8, HISTORICAL-WS 4, WS-S 2
Bank Marketing	14 of 30	NBI-S 17, NBI-R 7, HISTORICAL-WS 3, WS-S 2, HISTORICAL-WS/WS-S (tied) 1	NBI-S 15, WS-S 7, HISTORICAL-WS 4, NBI-R 3, HISTORICAL-WS/WS-S/NBI-S/NBI-R (tied) 1

The three identifying contrasts under both cost definitions: median paired difference in the CORE-relative hypervolume ratio (second minus first), with W/T/L, ties being differences within 10^{-12}; descriptive and untested. Under the leaf-count definition the values are those of the primary analysis; with its exact-zero tie rule Spambase HISTORICAL-WS $\rightarrow$ WS-S reads 16/7/7 rather than 16/8/6, one replication differing by 9.7×10^{-15}. Under measured time the geometry contrast stays positive on every dataset, smaller on Adult and Bank Marketing, and the anchor-provenance contrast changes sign on Adult.

dataset	contrast	leaf-count cost: median (W/T/L)	time cost: median (W/T/L)
MAGIC	HISTORICAL-WS → WS-S	+0.0000 (14/3/13)	-0.0000 (14/0/16)
MAGIC	WS-S → NBI-S	+0.0374 (26/0/4)	+0.0555 (23/0/7)
MAGIC	NBI-S → NBI-R	-0.0741 (2/0/28)	-0.0249 (7/0/23)
Spambase	HISTORICAL-WS → WS-S	+0.0018 (16/8/6)	+0.0060 (15/6/9)
Spambase	WS-S → NBI-S	+0.0500 (18/2/10)	+0.1875 (21/2/7)
Spambase	NBI-S → NBI-R	-0.6932 (5/0/25)	-0.9574 (7/3/20)
Adult	HISTORICAL-WS → WS-S	-0.0006 (9/4/17)	-0.0010 (13/2/15)
Adult	WS-S → NBI-S	+0.2549 (30/0/0)	+0.1195 (22/0/8)
Adult	NBI-S → NBI-R	-0.0192 (11/0/19)	+0.0696 (18/0/12)
Bank Marketing	HISTORICAL-WS → WS-S	+0.0000 (8/8/14)	+0.0030 (16/4/10)
Bank Marketing	WS-S → NBI-S	+0.2281 (24/0/6)	+0.0616 (19/1/10)
Bank Marketing	NBI-S → NBI-R	-0.4093 (7/0/23)	-0.4004 (5/1/24)

S5. Panel screening and dataset roles

dataset	C1 raw conflict	C2 front / curvature	C3 min R^2	C4 range	role
MAGIC	-0.2509	7 / 0.1729	0.6913	2706.3×	`primary_geometry_confirmatory`
Spambase	-0.4621	2 / undefined	0.8987	591.2×	`boundary_geometry_control`
Adult	-0.3977	6 / 0.1992	0.9078	3421.3×	`primary_geometry_confirmatory`
Bank Marketing	-0.4100	6 / 0.1983	0.8700	4355.5×	`primary_geometry_confirmatory`

Roles were assigned from these pre-campaign measurements and never revised from an optimizer outcome. They were measured on one partition per dataset, the pilot Stage A partition (seed 20260913), whose design-row evaluations are also the fitting sample of the frozen factor models (S4.1); it is not one of the 30 confirmatory partitions. Spambase was retained as a boundary geometry control by a pre-campaign amendment adopted after its criterion-2 failure was known (amendment 22): the replacement rule frozen at protocol v1, before the pilot, required a dataset failing criterion 1 or 2 to be replaced from the registry, and retaining Spambase departs from that rule. Its boundary-control result is therefore conditional on that screen and that amendment, not a prospective test of a hypothesis about the dataset; likewise the primary panel's results are conditional on a screen made on one pilot partition.

Every confirmatory replication evaluates the same 88 design rows on its own partition. The number of those rows that are non-dominated under the frozen objectives is summarized below as a post-campaign description; it changes no role.

dataset	non-dominated design rows, screening partition	confirmatory partitions: median	minimum	partitions with exactly two
MAGIC	7	9	6	0 of 30
Spambase	2	3	2	9 of 30
Adult	6	6	4	0 of 30
Bank Marketing	6	5	3	0 of 30

S6. Method and control registry

identifier	kind	runner stage	standalone budget	revalidated	isolates
HISTORICAL-WS-asrun	arm	historical_ws_asrun	108	yes	the frozen v0.1.0-dissertation solver, called unmodified, with the archived maximization orientation, observed-extrema normalization and asymmetric grid with no pure-quality vertex (20 points at the solver's default step), and no gate; its uncoded natural-unit surfaces follow the archived specification but are fitted by this study's code to this study's objectives. Reproduced, not repaired.
HISTORICAL-WS	arm	historical_ws	186	yes	the same weighted sum, solver, symmetric grid and surrogates as WS-S, differing ONLY in the normalization reference. WS-S to HISTORICAL-WS is therefore a single-factor contrast on normalization.
WS-S	arm	ws_s	186	yes	weighted-sum scalarization over the surrogate payoff reference. The geometry baseline for the primary contrast.
NBI-S	arm	nbi_s	186	yes	Normal Boundary Intersection on the same surrogates and the same surrogate reference as WS-S. WS-S to NBI-S isolates front-construction geometry.
NBI-R	arm	nbi_r	386	yes	NBI with anchors and payoff matrix from direct search on the REAL objectives. NBI-S to NBI-R isolates anchor and payoff provenance.
ANCHOR-INJECTION-CONTROL	control	anchor_injection_control	n/a	yes	NBI-S's own revalidated candidate set augmented with the same empirical anchors NBI-R receives, changing nothing else. It measures the indicator gain from adding those anchors alone to NBI-S's set; it was frozen to separate the set-composition part of the NBI-S to NBI-R gap from the relocated-geometry part, a reading that holds only if NBI-R returns its own anchors. Costs no new real evaluations.
GRID	baseline	direct_baselines	n/a	n/a	direct-search comparator at the frozen 386-evaluation comparator budget, which is matched to the most expensive arm (NBI-R) and not pairwise to each arm

identifier	kind	runner stage	standalone budget	revalidated	isolates
RANDOM	baseline	direct_baselines	n/a	n/a	direct-search comparator at the frozen 386-evaluation comparator budget, matched to the most expensive arm and not pairwise to each arm
BAYES-QUALITY	baseline	direct_baselines	n/a	n/a	single-objective Bayesian optimization on quality, at the frozen comparator budget; excluded from front indicators and from the AUGMENTED reference
BAYES-COST	baseline	direct_baselines	n/a	n/a	single-objective Bayesian optimization on cost; same exclusion
TPE-QUALITY	baseline	direct_baselines	n/a	n/a	single-objective TPE on quality; same exclusion
TPE-COST	baseline	direct_baselines	n/a	n/a	single-objective TPE on cost; same exclusion
NSGA2-MATCHED	baseline	direct_baselines	n/a	n/a	NSGA-II at 32 x 12 = 384 evaluations, run at the frozen comparator budget to within a disclosed two-evaluation shortfall (384 against 386); this is a budget-level match to the most expensive arm, not a pairwise match to NBI-S
NSGA2-UNMATCHED	context	nsga2_unmatched	n/a	n/a	NSGA-II at 10x the matched budget, one replication per dataset. A CONTEXT baseline: no fairness claim attaches to it and it enters no budget-matched comparison.
DESIGN	shared stage	design	n/a	n/a	88 face-centred central composite runs
EXTERNAL-VALIDATION-AUDIT	shared stage, AUDIT-ONLY	external_validation	n/a	n/a	78 points: the design's complementary half fraction plus 14 axial runs at half radius. Reaches the reliability gate diagnostics and NOTHING else. A gate failure changes no execution.
SURROGATE-ANCHORS	reference construction	surrogate_anchors	n/a	n/a	per-objective minimization of the fitted surrogates over the coded box
EMPIRICAL-ANCHORS	reference construction	empirical_anchors	n/a	n/a	direct search on the REAL objectives, one budget per objective. Best found within the declared budget; never described as certified optima.
REFERENCE-CORE	reference construction	reference_core	n/a	n/a	the reference front used for indicators
AUGMENTED-REFERENCE	reference construction	augmented_reference	n/a	n/a	reports self_grading_share_of_front
HOLDOUT-CONFIRMATION	confirmation stage	holdout_confirmation	n/a	n/a	selected candidates re-measured on the held-out partition

n/a: not applicable. Standalone budgets are defined for the five arms; the direct-search comparators run at the frozen comparator budget of 386 real evaluations (matched NSGA-II, 384), and the unmatched NSGA-II context run at 10x the matched budget. Revalidation applies to the arms, whose candidates are selected on the surrogates, and to the control, which reuses NBI-S's revalidated set; the other entities evaluate on the real learner directly or return no candidate set.

S7. Budget

method	stage	logical per unit
anchor_injection_control	candidate_validation	2
bayes_cost	direct_search	386
bayes_quality	direct_search	386
design	design	88
empirical_anchor_search	anchor	200
external_validation_audit	external_audit	78
grid	direct_search	386
historical_ws_asrun_revalidation	candidate_validation	20
historical_ws_revalidation	candidate_validation	20
holdout_confirmation	holdout_audit	5
nbi_r_revalidation	candidate_validation	20
nbi_s_revalidation	candidate_validation	20
nsga2	direct_search	384
random	direct_search	386
tpe_cost	direct_search	386
tpe_quality	direct_search	386
ws_s_revalidation	candidate_validation	20
total		**3,173**

Of which 83 audit-only and 3,090 solution-producing.

quantity	value
logical evaluations per unit	3,173
confirmatory units	120
unmatched NSGA-II, outside the unit	15,360
campaign total	**396,120**
of which audit-only	9,960
unique physical evaluations	377,316

The campaign total already contains the unmatched NSGA-II charge, inside the four replication-0 ledgers. Adding it again gives 411,480 and is wrong.

S7.1 Standalone cost per arm

arm	standalone logical
HISTORICAL-WS-asrun	108
HISTORICAL-WS	186
WS-S	186
NBI-S	186
NBI-R	386
each direct-search comparator except NSGA-II	386
NSGA-II, matched (population 32, 12 generations)	384

The comparator budget is the maximum over arms (NBI-R). NSGA-II's frozen population and generation count give 384 real evaluations, 2 fewer than that budget. A direct-search comparator therefore receives about twice the real evaluations WS-S and NBI-S require standalone, and any comparison against them must say so.

S7.2 The grid at 186 evaluations (retrospective, post hoc)

The grid's stored rows are in evaluation order, and the frozen code fixes that order: the 128-point two-level corner mesh first, then seeded uniform padding. In every unit the first 186 stored rows are exactly the grid the frozen code builds at a budget of 186 with the same seed, so they measure, with the campaign's own evaluations, the grid the protocol would have run at WS-S's and NBI-S's standalone budget. They are scored with the frozen indicator code against each unit's stored CORE front, after the full 386 rows have been rescored and found equal to the stored values and Table 5 reproduced. No grid point was evaluated (`scripts/grid_budget_prefix.py`, `analysis/grid_budget_prefix.json`).

dataset	grid evaluations	GRID median	median NBI-S − GRID	95% interval	NBI-S / GRID wins
MAGIC	386	1.0068	-0.0628	[-0.1122, -0.0480]	0 / 30
MAGIC	186	0.9996	-0.0504	[-0.0829, -0.0437]	0 / 30
Spambase	386	1.0192	-0.1286	[-0.3472, +0.1511]	12 / 18
Spambase	186	1.0192	-0.1286	[-0.3472, +0.1511]	12 / 18
Adult	386	1.0476	-0.0488	[-0.0783, -0.0245]	5 / 25
Adult	186	1.0461	-0.0488	[-0.0783, -0.0245]	5 / 25
Bank Marketing	386	1.0625	-0.1323	[-0.2891, -0.0182]	10 / 20
Bank Marketing	186	1.0625	-0.1323	[-0.2888, -0.0182]	10 / 20

Stopped at 186 evaluations the grid keeps its lead: its wins against NBI-S are unchanged on Spambase, Adult and Bank Marketing and stay at 30 of 30 on MAGIC, so the frozen 21-of-30 direction rule is met on MAGIC and Adult at either budget. Almost all of the grid's advantage comes from its first 186 points. Those include the 64 corners that re-measure design configurations inside CORE at both budgets, a larger share at 186 (34%) than at 386 (17%); this is reported, not corrected. The prefix is not a budget-matched comparator arm: the arms were not re-run, and the budget rule of Section 8.2 is unchanged.

S8. Randomness

Seeds derive from BLAKE2b over the namespace `xgboost-hpo-vrfnbi/v3` and the tuple (dataset, replication, method, stage), fed to `numpy.random.SeedSequence`.

dataset	unit seed, replication 0	example method seed
MAGIC	20261914	12501833259513610416
Spambase	20262914	1639200741726807977
Adult	20263914	3386437749439664214
Bank Marketing	20264914	7633668092872975921

All 3,840 campaign streams are distinct and stable across processes and `PYTHONHASHSEED`.

S9. Primary statistics, both references

S9.1 CORE reference (primary)

dataset	contrast	median Δ	95% CI	W/T/L	rank-biserial	Holm p	Nadeau–Bengio p
MAGIC	HISTORICAL-WS -> WS-S	+0.0000	[-0.0022, +0.0056]	14/3/13	+0.14	0.5165	0.6697
MAGIC	WS-S -> NBI-S	+0.0374	[+0.0144, +0.0687]	26/0/4	+0.92	9.425e-07	0.0860
MAGIC	NBI-S -> NBI-R	-0.0741	[-0.0957, -0.0482]	2/0/28	-0.99	2.794e-08	0.0254
Adult	HISTORICAL-WS -> WS-S	-0.0006	[-0.0033, +0.0000]	9/4/17	-0.28	0.2087	0.7467
Adult	WS-S -> NBI-S	+0.2549	[+0.1740, +0.5614]	30/0/0	+1.00	5.588e-09	0.0148
Adult	NBI-S -> NBI-R	-0.0192	[-0.1610, +0.0299]	11/0/19	-0.37	0.1607	0.4161
Bank Marketing	HISTORICAL-WS -> WS-S	+0.0000	[-0.0088, +0.0000]	8/8/14	-0.21	0.3896	0.8893
Bank Marketing	WS-S -> NBI-S	+0.2281	[+0.1670, +0.4741]	24/0/6	+0.88	9.717e-06	0.0764
Bank Marketing	NBI-S -> NBI-R	-0.4093	[-0.5821, -0.2059]	7/0/23	-0.82	4.733e-05	0.0791
Spambase	HISTORICAL-WS -> WS-S	+0.0018	[+0.0000, +0.0398]	16/7/7	+0.41	0.1175	0.6846
Spambase	WS-S -> NBI-S	+0.0500	[-0.0043, +0.4369]	18/2/10	+0.41	0.1175	0.6550
Spambase	NBI-S -> NBI-R	-0.6932	[-0.9839, -0.5217]	5/0/25	-0.78	0.0005664	0.2763

The Holm p column is the secondary test: a two-sided Wilcoxon signed-rank test, Holm-adjusted within each dataset family (Spambase forms its own family). The descriptive triple on the CORE reference (median Δ with its 95% percentile bootstrap interval, W/T/L and rank-biserial) is the primary evidence; the AUGMENTED reference is a pre-declared sensitivity. The Nadeau–Bengio column is the pre-declared corrected resampled t, at an inflation of $\sqrt{8.5} = 2.9155$, not adjusted for multiplicity. It is a sensitivity, not the study's significance test, and is reported in full including where it is adverse to the finding.

S9.2 AUGMENTED reference (mandatory sensitivity)

dataset	contrast	median Δ	95% CI	W/T/L	rank-biserial	Holm p	Nadeau–Bengio p
MAGIC	HISTORICAL-WS -> WS-S	+0.0000	[-0.0029, +0.0047]	13/3/14	+0.05	0.8288	0.7807
MAGIC	WS-S -> NBI-S	+0.0382	[+0.0179, +0.0668]	27/0/3	+0.95	2.049e-07	0.0927
MAGIC	NBI-S -> NBI-R	-0.0794	[-0.0939, -0.0451]	1/0/29	-0.99	2.794e-08	0.0286
Adult	HISTORICAL-WS -> WS-S	-0.0006	[-0.0018, +0.0000]	8/4/18	-0.23	0.3158	0.7431
Adult	WS-S -> NBI-S	+0.1936	[+0.1322, +0.3717]	30/0/0	+1.00	5.588e-09	0.0204
Adult	NBI-S -> NBI-R	-0.0535	[-0.1181, +0.0035]	10/0/20	-0.54	0.01741	0.3756
Bank Marketing	HISTORICAL-WS -> WS-S	-0.0001	[-0.0021, +0.0000]	7/8/15	-0.23	0.3382	0.7957
Bank Marketing	WS-S -> NBI-S	+0.1792	[+0.1074, +0.2045]	25/0/5	+0.94	7.655e-07	0.0643
Bank Marketing	NBI-S -> NBI-R	-0.1699	[-0.2369, -0.0772]	7/0/23	-0.82	4.733e-05	0.1109
Spambase	HISTORICAL-WS -> WS-S	+0.0000	[+0.0000, +0.0096]	14/11/5	+0.48	0.1283	0.6545
Spambase	WS-S -> NBI-S	+0.0503	[+0.0000, +0.2079]	19/2/9	+0.33	0.1283	0.6531
Spambase	NBI-S -> NBI-R	-0.3464	[-0.4880, -0.2553]	4/1/25	-0.84	0.0002382	0.0658

The Holm p column is the secondary test: a two-sided Wilcoxon signed-rank test, Holm-adjusted within each dataset family (Spambase forms its own family). The descriptive triple on the CORE reference (median Δ with its 95% percentile bootstrap interval, W/T/L and rank-biserial) is the primary evidence; the AUGMENTED reference is a pre-declared sensitivity. The Nadeau–Bengio column is the pre-declared corrected resampled t, at an inflation of $\sqrt{8.5} = 2.9155$, not adjusted for multiplicity. It is a sensitivity, not the study's significance test, and is reported in full including where it is adverse to the finding.

Self-grading share of the AUGMENTED front, which the frozen protocol requires beside every AUGMENTED contrast; each contrast above compares two of the methods tabulated here. The share is the fraction of the AUGMENTED reference's non-dominated points contributed by the points a method is scored on (an arm's revalidated candidates, a comparator's evaluations), median over the 30 replications, from `analysis/indicators_long.csv`. A front point contributed by more than one method counts once for each of them. The anchor-injection control contributes no candidates of its own to the reference.

dataset	HISTORICAL-WS-asrun	HISTORICAL-WS	WS-S	NBI-S	NBI-R	GRID	RANDOM	NSGA2-MATCHED
MAGIC	0.000	0.055	0.058	0.104	0.080	0.182	0.020	0.335
Spambase	0.000	0.036	0.080	0.200	0.000	0.121	0.000	0.000
Adult	0.000	0.070	0.069	0.135	0.132	0.212	0.000	0.180
Bank Marketing	0.000	0.093	0.059	0.121	0.018	0.204	0.000	0.214

S10. Secondary indicators

dataset	entity	HV ratio	IGD$^+$	GD	spacing	joint-ND	front size
MAGIC	HISTORICAL-WS-asrun	0.7627	0.1477	0.1174	0.0460	0.1458	8.0000
MAGIC	HISTORICAL-WS	0.8613	0.1028	0.1153	0.0890	0.5798	10.0000
MAGIC	WS-S	0.8614	0.0930	0.1013	0.0862	0.5857	9.0000
MAGIC	NBI-S	0.9430	0.0511	0.0781	0.0496	0.5917	11.0000
MAGIC	NBI-R	0.8588	0.0856	0.1015	0.0244	0.6667	9.0000
MAGIC	ANCHOR-INJECTION-CONTROL	0.9607	0.0444	0.0713	0.0979	0.6259	12.5000
MAGIC	GRID	1.0068	0.0036	0.0393	0.0706	0.9181	17.0000
MAGIC	RANDOM	0.8907	0.0629	0.0775	0.0382	0.4416	13.0000
MAGIC	NSGA2-MATCHED	0.9580	0.0373	0.0858	0.0188	0.6883	30.5000
Spambase	HISTORICAL-WS-asrun	0.0221	0.6561	0.7892	0.1642	0.0000	4.0000
Spambase	HISTORICAL-WS	0.8583	0.1312	0.1837	0.1445	0.8750	3.0000
Spambase	WS-S	0.8819	0.1054	0.2042	0.2049	1.0000	3.0000
Spambase	NBI-S	1.0812	0.0877	0.2343	0.2142	0.9444	4.0000
Spambase	NBI-R	0.0793	0.6200	0.5881	0.0000	1.0000	3.0000
Spambase	ANCHOR-INJECTION-CONTROL	1.0812	0.0877	0.2343	0.2142	0.9444	4.0000
Spambase	GRID	1.0192	0.0000	0.0394	0.2372	1.0000	5.0000
Spambase	RANDOM	0.0000	1.3453	1.3795	0.0256	0.0000	2.0000
Spambase	NSGA2-MATCHED	0.4164	0.4738	0.5878	0.1166	0.2500	5.0000
Adult	HISTORICAL-WS-asrun	0.5196	0.3404	0.2414	0.0176	0.0000	10.5000
Adult	HISTORICAL-WS	0.5422	0.1847	0.1483	0.0930	0.6970	7.5000
Adult	WS-S	0.6028	0.1688	0.1785	0.0890	0.7321	7.0000
Adult	NBI-S	0.9951	0.0579	0.1077	0.1329	0.8258	8.5000
Adult	NBI-R	0.9098	0.1201	0.1374	0.0503	0.7071	9.0000
Adult	ANCHOR-INJECTION-CONTROL	0.9951	0.0578	0.1397	0.1483	0.7386	10.0000
Adult	GRID	1.0476	0.0000	0.0465	0.0947	1.0000	8.5000
Adult	RANDOM	0.4369	0.3334	0.2441	0.0984	0.0000	5.0000
Adult	NSGA2-MATCHED	0.8650	0.1228	0.1821	0.0359	0.4715	28.0000
Bank Marketing	HISTORICAL-WS-asrun	0.2743	0.4275	0.2907	0.0784	0.0000	6.0000
Bank Marketing	HISTORICAL-WS	0.6082	0.1946	0.2087	0.1804	0.6333	7.0000
Bank Marketing	WS-S	0.6010	0.1948	0.2320	0.1862	0.5000	7.0000
Bank Marketing	NBI-S	0.9979	0.0915	0.2173	0.1219	0.6111	9.0000
Bank Marketing	NBI-R	0.5330	0.3582	0.2468	0.0505	0.2917	6.0000
Bank Marketing	ANCHOR-INJECTION-CONTROL	0.9979	0.0898	0.1754	0.1358	0.6667	8.5000
Bank Marketing	GRID	1.0625	0.0000	0.0481	0.1432	1.0000	8.0000
Bank Marketing	RANDOM	0.3190	0.3257	0.3119	0.1037	0.0000	7.0000
Bank Marketing	NSGA2-MATCHED	0.9674	0.0934	0.2182	0.0451	0.4929	20.5000

Undefined spacing. 140 cells are non-finite: spacing and spacing_cv on the 70 method-by-reference blocks whose front has one point. Schott spacing requires at least two gaps, so NaN is the correct value and those units are excluded from spacing summaries rather than propagated.

S11. Surrogate-gate regimes

dataset	quality gate	cost gate	both	role
MAGIC	8/30	100%	27%	primary_geometry_confirmatory
Spambase	9/30	100%	30%	boundary_geometry_control
Adult	0/30	73%	0%	primary_geometry_confirmatory
Bank Marketing	0/30	90%	0%	primary_geometry_confirmatory

The gate is diagnostic, never adaptive: every arm ran at every replication whatever the gate said.

Every primary contrast by composite-quality gate status, from `analysis/indicators_long.csv`: median paired difference (second minus first) in the CORE-relative hypervolume ratio, with W/T/L, descriptive and untested. The cost gate passed in every MAGIC and Spambase replication, so these are also the strata of the joint gate. No Adult or Bank Marketing replication passed, so for those datasets the failing stratum is the whole sample of S9.1; n/a marks an empty stratum.

dataset	contrast	gate passed: n, median difference, W/T/L	gate failed: n, median difference, W/T/L
MAGIC	HISTORICAL-WS -> WS-S	8, -0.0031, 2/1/5	22, +0.0018, 12/2/8
MAGIC	WS-S -> NBI-S	8, +0.1173, 8/0/0	22, +0.0165, 18/0/4
MAGIC	NBI-S -> NBI-R	8, -0.0577, 1/0/7	22, -0.0901, 1/0/21
Spambase (boundary)	HISTORICAL-WS -> WS-S	9, +0.0000, 3/4/2	21, +0.0173, 13/3/5
Spambase (boundary)	WS-S -> NBI-S	9, +0.0000, 4/1/4	21, +0.0641, 14/1/6
Spambase (boundary)	NBI-S -> NBI-R	9, -0.5238, 3/0/6	21, -0.8770, 2/0/19
Adult	HISTORICAL-WS -> WS-S	0, n/a, n/a	30, -0.0006, 9/4/17
Adult	WS-S -> NBI-S	0, n/a, n/a	30, +0.2549, 30/0/0
Adult	NBI-S -> NBI-R	0, n/a, n/a	30, -0.0192, 11/0/19
Bank Marketing	HISTORICAL-WS -> WS-S	0, n/a, n/a	30, +0.0000, 8/8/14
Bank Marketing	WS-S -> NBI-S	0, n/a, n/a	30, +0.2281, 24/0/6
Bank Marketing	NBI-S -> NBI-R	0, n/a, n/a	30, -0.4093, 7/0/23

S12. Controls

S12.1 Anchor-injection control

dataset	injection effect	full NBI-S→NBI-R gap	CHIM extent ratio (post hoc)
MAGIC	+0.0133	-0.0741	0.811
Spambase	+0.0000	-0.6932	0.370
Adult	+0.0000	-0.0192	0.204
Bank Marketing	+0.0000	-0.4093	0.272

The control set is a superset of NBI-S's, so its effect is non-negative by construction. The CHIM extent ratio is post hoc: it was first computed after the confirmatory results, from the recorded payoff matrices, and is not part of the frozen protocol. Within-dataset Spearman between the per-replication CHIM extent ratio and the hypervolume gap: MAGIC -0.11, Spambase +0.32, Adult +0.68, Bank Marketing +0.68. A post hoc association, not a cause: clear on Adult and Bank Marketing, weak on Spambase and absent on MAGIC.

S12.2 Solver health, NBI-S against NBI-R

dataset	certified (S/R)	max equality residual (S/R)	max rounding displacement (S/R)	distinct configs (S/R)
MAGIC	1.000 / 1.000	8.6e-10 / 9.3e-10	2.8e-02 / 5.2e-03	20 / 20
Spambase	1.000 / 1.000	5.9e-10 / 5.8e-10	1.7e-03 / 1.7e-12	20 / 20
Adult	1.000 / 1.000	4.4e-10 / 5.6e-10	2.7e-02 / 8.3e-04	20 / 20
Bank Marketing	1.000 / 1.000	6.1e-10 / 6.8e-10	8.2e-03 / 9.5e-04	20 / 20

Solver behaviour was comparable between `NBI-S` and `NBI-R` and does not explain the deficit. Across the 240 arm-units the certified fraction had median 1.000, with two exceptions at 0.900 and 0.950; the per-unit maximum equality residual had median 6.6e-10, against a campaign maximum of 6.9e-1 on a single MAGIC `NBI-S` unit; and both arms returned 20 distinct realized configurations in 239 of the 240 arm-units, with 19 in that same MAGIC `NBI-S` unit. The table entries are per-dataset medians of per-unit values; the rounding-displacement column is, per unit, the largest objective displacement induced by rounding on the fitted surfaces. Solver behaviour is therefore comparable, but not identical, and the deficit is not solver failure or candidate collapse. The protocol's fixed-integer re-solve was not implemented (Section 8.7), so the rounded points actually evaluated carry no equality certificate.

S12.3 Historical reconstruction

dataset	as-run	shared specification	WS-S	Δ(shared − as-run)
MAGIC	0.7627	0.8613	0.8614	+0.0994
Spambase	0.0221	0.8583	0.8819	+0.5404
Adult	0.5196	0.5422	0.6028	+0.1480
Bank Marketing	0.2743	0.6082	0.6010	+0.3126

S13. Baselines

dataset	GRID	RANDOM	NSGA-II matched	WS-S	NBI-S	NBI-S wins vs GRID
MAGIC	1.0068	0.8907	0.9580	0.8614	0.9430	0/30
Spambase	1.0192	0.0000	0.4164	0.8819	1.0812	12/30
Adult	1.0476	0.4369	0.8650	0.6028	0.9951	5/30
Bank Marketing	1.0625	0.3190	0.9674	0.6010	0.9979	10/30

The unmatched NSGA-II run receives ten times the matched budget on one replication per dataset and is a context baseline: no fairness claim attaches to it and it enters no budget-matched comparison and neither reference.

S13.1 Single-objective Bayesian-optimization and TPE runs

Declared in the frozen protocol (section 3): the four single-objective runs produce single-objective optima, not fronts, so they are reported on their own endpoints only, labelled as single-objective runs, and enter neither the front-indicator tables nor either reference. Computed after the campaign from stored outputs by `scripts/single_objective_endpoints.py`. Each run used the comparator budget of 386 real evaluations per

replication. The endpoint is the best value of the run's own objective (f_{quality} or f_{cost}, both minimized) among its real evaluations. For context only, the last column gives the median best value of the same objective in each replication's CORE reference (88 design rows and 200 anchor-search rows), to which no compared method contributes; that comparison was not pre-specified, is descriptive and carries no test.

dataset	run	objective	best value: median (interquartile range)	CORE best value: median
MAGIC	BAYES-QUALITY	f_{quality}	-0.810 (-0.826 to -0.798)	-0.735
MAGIC	BAYES-COST	f_{cost}	-1.800 (-1.847 to -1.785)	-2.052
MAGIC	TPE-QUALITY	f_{quality}	-0.805 (-0.822 to -0.791)	-0.735
MAGIC	TPE-COST	f_{cost}	-1.794 (-1.850 to -1.751)	-2.052
Spambase	BAYES-QUALITY	f_{quality}	-0.855 (-0.872 to -0.819)	-0.845
Spambase	BAYES-COST	f_{cost}	-2.586 (-2.710 to -2.363)	-3.028
Spambase	TPE-QUALITY	f_{quality}	-0.816 (-0.848 to -0.768)	-0.845
Spambase	TPE-COST	f_{cost}	-2.310 (-2.515 to -2.170)	-3.028
Adult	BAYES-QUALITY	f_{quality}	-0.707 (-0.712 to -0.700)	-0.683
Adult	BAYES-COST	f_{cost}	-2.237 (-2.310 to -2.188)	-2.387
Adult	TPE-QUALITY	f_{quality}	-0.706 (-0.713 to -0.700)	-0.683
Adult	TPE-COST	f_{cost}	-2.163 (-2.238 to -2.010)	-2.387
Bank Marketing	BAYES-QUALITY	f_{quality}	-0.930 (-0.946 to -0.910)	-0.887
Bank Marketing	BAYES-COST	f_{cost}	-2.583 (-2.639 to -2.457)	-2.516
Bank Marketing	TPE-QUALITY	f_{quality}	-0.919 (-0.934 to -0.909)	-0.887
Bank Marketing	TPE-COST	f_{cost}	-2.402 (-2.530 to -2.207)	-2.516

S13.2 Unmatched NSGA-II run

Declared in `protocol/NSGA2_DECISION.md` (section 5): the unmatched run, ten times the matched budget on one replication per dataset, asks only whether a weak matched result is explained by budget starvation. It is not an efficiency comparison, no claim about any arm is made relative to it, and it is reported in its own table with the generation count stated. Computed after the campaign from stored outputs by `scripts/nsga2_unmatched_table.py`, which scores the stored rows of the run against that replication's CORE reference with the frozen indicator code.

dataset	replication	real evaluations	population	generations	CORE-relative HV ratio, unmatched	matched NSGA-II, same replication	non-dominated rows (distinct points)
MAGIC	0	3,840	32	120	1.0269	0.9357	186 (105)
Spambase	0	3,840	32	120	6.4459	0.0937	1091 (403)
Adult	0	3,840	32	120	1.4053	0.8060	176 (108)
Bank Marketing	0	3,840	32	120	2.0001	0.7862	177 (72)

The unmatched budget is ten times the realized matched budget, 3,840 real evaluations (ten times 384), as fixed in `protocol/NSGA2_DECISION.md` (section 5) and executed by the runner; the protocol's general section states 3,860 (ten times 386). Non-dominated rows count every non-dominated stored row, so repeated configurations of the converged population are counted separately; the distinct points are in parentheses. Spambase's CORE front has only three points in this replication, so its normalization box is narrow and the hypervolume ratio is unbounded above (S15); its magnitude is not comparable with the other datasets'.

S14. Holdout confirmation

dataset	arm	internal	holdout	median paired drop
MAGIC	historical_ws	0.8580	0.8604	-0.0017
MAGIC	historical_ws_asrun	0.8477	0.8541	-0.0034
MAGIC	nbi_r	0.8587	0.8603	-0.0013
MAGIC	nbi_s	0.8593	0.8594	-0.0015
MAGIC	ws_s	0.8598	0.8628	-0.0016
Spambase	historical_ws	0.8871	0.8947	-0.0059
Spambase	historical_ws_asrun	0.9046	0.9088	-0.0044
Spambase	nbi_r	0.9185	0.9235	-0.0117
Spambase	nbi_s	0.8898	0.9072	-0.0110
Spambase	ws_s	0.8890	0.8990	-0.0076
Adult	historical_ws	0.8408	0.8392	+0.0016
Adult	historical_ws_asrun	0.8412	0.8406	+0.0004
Adult	nbi_r	0.8433	0.8430	-0.0018
Adult	nbi_s	0.8437	0.8445	-0.0012
Adult	ws_s	0.8409	0.8415	+0.0020
Bank Marketing	historical_ws	0.8985	0.8970	+0.0009
Bank Marketing	historical_ws_asrun	0.8975	0.8974	+0.0011
Bank Marketing	nbi_r	0.8984	0.8990	-0.0002
Bank Marketing	nbi_s	0.8986	0.8973	+0.0012
Bank Marketing	ws_s	0.8988	0.8973	+0.0014

All magnitudes are below 0.012 and several are negative, meaning the held-out partition scored better than the internal resampling. This is read as the absence of gross selection optimism, **not** as a ranking of arms.

S15. Finite-reference diagnostics

dataset	rows with HV ratio > 1	share	maximum
MAGIC	36/270	13%	1.057
Spambase	95/270	35%	9.912
Adult	76/270	28%	1.176
Bank Marketing	93/270	34%	2.349

Overall 27.8% of rows exceed 1, maximum 9.912. The CORE reference is a finite method-independent set of 288 points, not the true Pareto front; a ratio above 1 means the candidate set improved on that finite reference.

S16. Claims and evidence

The manuscript's headline numerical claims, each with its estimand, source artifact, verification status, allowed wording and prohibited stronger wording, are in `CONFIRMATORY_CLAIMS_AND_EVIDENCE.md`. Two are recorded in corrected form because a post-campaign recomputation from the raw unit artifacts refuted the author's version. The ledger predates the coauthor reviews; where its allowed wording differs from the manuscript's, as for the post hoc status of the signed-rank tests and the CHIM-contraction association, the manuscript's wording supersedes it, and the grid at 186 evaluations (S7.2) and the centre-point sensitivity (S18) are reported only here.

S17. Reproducibility manifest

item	value
manuscript source commit	`8780c72828d8780d0ceb3fc892547670c7b8a560`
package tag	not yet created; planned name `paper2-manuscript-v6`, to be created after the authors' approval
relationship	planned: the commit to be tagged will differ from its manuscript source commit only by provenance metadata and compiled outputs
protocol tag	`xgboost-hpo-protocol-v3`
results tag	`xgboost-hpo-confirmatory-results-v1`
datasets	MAGIC, Spambase, Adult, Bank Marketing
replications	30 per dataset, 120 units
primary endpoint	hv_ratio against the CORE reference
campaign runtime	9 h 46 min, 14 workers × 1 thread

What is public and what is not. The repository is public. Its code, protocol, amendment ledger, the design with its SHA-256, the frozen factor models, the per-unit indicator table and the confirmatory analysis are pinned by the tags of S1; the per-unit NBI mechanism tables and the analyses computed after the campaign or in coauthor review, with their scripts, are on the revision branch and will be pinned by the planned manuscript-package tag. The four datasets are not redistributed: they are public UCI and OpenML mirrors under CC BY 4.0, obtained independently and checked against the recorded SHA-256 on acquisition. The per-unit experiment tree is not versioned and is not yet deposited: 120 unit directories of stage checkpoints and one SQLite evaluation cache each, about 0.8 GB, holding configurations, seeds and evaluated responses but no data rows. Every aggregate in the paper is stored in a committed artifact, and those of the primary analysis can be recomputed from the committed per-unit tables; regenerating those tables, and the post-campaign diagnostics that read stored rows (Section 8.7, S7.2, S18), needs that tree, and re-running a unit needs the datasets, the code at the protocol tag and the seeds of S8. Exact reproduction across platforms or linear-algebra backends was not tested (Section 9.4).

S18. Centre-point sensitivity: 88 design rows against 79 distinct configurations (post hoc)

The design's ten centre rows are one configuration. In every confirmatory unit the evaluation cache computed it once and served it nine times (88 design requests, 79 distinct keys); in the Stage A pilot that fixed the factor models the ten were executed separately and returned identical values of all seven responses, only the measured time differing. They therefore carry no pure-error information, yet the 88-row fits weight the centre tenfold and count all 88 rows in the elimination's t-tests (Section 5.6), and the factor models weight it tenfold in their means, standard deviations and components.

The diagnostic (`scripts/center_point_sensitivity.py`, `analysis/center_point_sensitivity.json`) uses stored responses only. It first reproduces every stored 88-row surrogate fit (terms and coefficients), every stored external-audit score and gate decision and every stored surrogate anchor exactly; it then refits the same procedure on the 79 distinct configurations, keeping one centre row, with the frozen factor model and both objectives unchanged; and, separately, it refits each dataset's factor model on the 79 distinct pilot configurations. No learner was trained, no candidate evaluated and no arm re-run; anchors were minimized on the polynomial surfaces with the frozen multistart. The comparison criteria and the classification rule were committed before the first run on the campaign records and are stored in the artifact.

dataset	objective	terms identical	within 2 terms	prediction-stable	residual df, 88 → 79	gate passes, 88 → 79	anchor moved, refit	anchor moved, restart only
MAGIC	quality	10/30	25/30	18/30	70–77 → 62–69	8 → 1	19/30	17/30
MAGIC	cost	6/30	26/30	30/30	70–75 → 62–66	30 → 30	15/30	5/30
Spambase	quality	14/30	26/30	24/30	69–76 → 61–68	9 → 7	20/30	9/30
Spambase	cost	18/30	27/30	27/30	73–78 → 65–70	30 → 30	15/30	7/30
Adult	quality	16/30	29/30	28/30	75–78 → 67–69	0 → 0	18/30	26/30
Adult	cost	18/30	29/30	30/30	72–76 → 64–67	22 → 21	9/30	0/30
Bank Marketing	quality	23/30	29/30	29/30	77–81 → 70–72	0 → 0	23/30	28/30
Bank Marketing	cost	6/30	16/30	14/30	66–76 → 60–68	27 → 23	16/30	1/30

Prediction-stable: root-mean-square difference between the two surfaces over the 78 audit points, the 79 design points and 4,096 uniform points of the coded box at most a quarter of the 88-row surface's external RMSE, with Spearman at least 0.99 between the two prediction sets. Anchor moved: the objective's surrogate minimizer moves by more than 10% of the coded box diagonal or a payoff entry changes by more than 0.10 objective units. Restart only: the same anchor criterion between the stored anchors and the unchanged 88-row surfaces minimized from a different set of multistart points.

dataset	min Tucker congruence	roles and orientation	quality weights, frozen → 79 rows	min Spearman, frozen vs 79-row objectives	Kaiser count
MAGIC	0.9998	same	0.556, 0.444 → 0.544, 0.456	0.9995	2 → 2
Spambase	0.9999	same	0.775, 0.225 → 0.776, 0.224	0.9993	2 → 2
Adult	0.9998	same	0.547, 0.453 → 0.551, 0.449	0.9990	1 → 1
Bank Marketing	0.9990	same	0.587, 0.413 → 0.589, 0.411	0.9959	1 → 1

The factor models do not move: every factor keeps its role and orientation, congruence is at least 0.999, and the two objective definitions rank the pilot configurations and the pooled audit rows almost identically. The surfaces' predictions meet the pre-declared stability criterion for most surfaces but not for MAGIC's quality surface (18 of 30) or Bank Marketing's cost surface (14 of 30), whose selected terms also differ by three or more in 14 of 30 fits; elsewhere the terms differ by at most two in 25 or more of 30. At a given number of terms the residual degrees of freedom of the elimination are nine fewer. Gate decisions change most for MAGIC's quality surface (8 → 1 passes) and Bank Marketing's cost surface (27 → 23), and at most twice elsewhere; the composite-quality gate stays at 0 of 30 on Adult and Bank Marketing.

The anchor flags of the refit have three sources, counted below (descriptive, computed after the first run had returned the class): the minimizer moved to a point equally good on both surfaces (within 0.01 objective units), it moved to a distinct minimum, or it stayed where it was and a payoff value changed because the surface did.

dataset	objective	flagged by the refit	moved, equal minimum	moved, distinct minimum	payoff value only	flagged by a restart alone
MAGIC	quality	19/30	4	13	2	17/30
MAGIC	cost	15/30	3	5	7	5/30
Spambase	quality	20/30	11	8	1	9/30
Spambase	cost	15/30	10	4	1	7/30
Adult	quality	18/30	10	5	3	26/30
Adult	cost	9/30	0	0	9	0/30
Bank Marketing	quality	23/30	14	7	2	28/30
Bank Marketing	cost	16/30	1	4	11	1/30

The restart check, also added after the first run, minimizes the unchanged 88-row surfaces from a different set of multistart points. It moves the quality anchor in 17, 9, 26 and 28 of 30 replications on MAGIC, Spambase, Adult and Bank Marketing, in 79 of 80 cases to a point within 0.01 of the same minimum: the quality surfaces have flat sets of equally good minima, the mechanism stated in Section 5.6, and the frozen seeded multistart picks one of them. The check shares the minimizer's deterministic box-centre start, so it can only understate that variability, and it cannot produce payoff changes at an unmoved anchor. It accounts for much of the refit's quality-anchor movement on Adult and Bank Marketing, but not on MAGIC, where the refit moved the quality anchor to a distinct minimum in 13 of 30 replications, nor on Spambase, where a restart moved it in only 9. The cost anchor never moved on Adult: all its flags are payoff changes at an unchanged location, as are most of Bank Marketing's. Under the refit NBI-S's quasi-normal direction moves by at most 46.3, 32.7, 3.2 and 4.5 degrees, and the per-dataset medians of the CHIM-extent ratio (main text, Figure 3) change from 0.811, 0.370, 0.204 and 0.272 to 0.815, 0.273, 0.193 and 0.274, in the same dataset order.

Under the pre-declared rule the sensitivity is classified **HIGH** on every dataset, by the anchor criterion alone (anchors flagged in more than 15 of 30 replications for at least one objective); the factor and gate criteria stay below their high levels, while the prediction and term criteria alone would have made MAGIC, Spambase and Bank Marketing moderate. The diagnostic does not show whether any primary contrast changes: that would need the surrogate-based arms re-run on the alternative surfaces, or with a rule that chooses among equally good minima, and their candidates revalidated with new learner fits, which was not done. The confirmatory results are those of the frozen 88-row specification and of its seeded minimizer, and the claims that rest on the surrogate payoff geometry are qualified in the Abstract and in Sections 6.4, 6.4.1, 7.4, 7.5 and 8.5. The alternative fits replace nothing and rescore nothing.